europass



Asim Shahzad

Date of birth: 1995 | **Place of birth:** Pakistan | **Nationality:** Pakistani | **Gender:** Male |

Phone number: (+61) 424192211 (Mobile) | **Phone number:** (+92) 3000793014 (Home) |

Email address: asim.shahzad1993@gmail.com | **LinkedIn:**

https://www.linkedin.com/in/asim-shahzad95/ | **ORCID:** https://orcid.org/0009-0000-6948-3790 |

Whatsapp Messenger: +61424192211 | **WeChat:** 15668328215 |

Address: Birchgrove Cres, Eastwood, NSW, 2122, Sydney, Australia (Home)

ABOUT ME

I am a researcher with a strong interest in interdisciplinary work that integrates artificial intelligence, data-driven modeling, signal processing, and optimization for real-world engineering applications. My research focuses on developing advanced deep learning and intelligent decision-making frameworks for complex industrial systems, including sensor-based rechargeable wireless networks, signal analysis, predictive maintenance, fault diagnosis, uncertainty-aware decision-making, and battery-swapping station optimization. I am particularly motivated to combine theoretical rigor with practical impact, collaborate with leading researchers, and contribute to innovative solutions for intelligent and sustainable engineering systems.

EDUCATION AND TRAINING

18 NOV 2011 – 25 MAY 2016 Multan, Pakistan

COMPUTER ENGINEERING Bahauddin Zakariya University

Fundamental Subjects Studied: Programming, Computer Networks, Digital Logic Design, Circuit Analysis, Basic Electrical Enginnering, Electronics (I, II, III), Data Structure, Object Oriented Programming, Signal & Systems, Database & Software Engineering, Computer Communication & Networks, Digital Signal Processing, Microprocessor, AI & Robotics, Control Engineering, Embedded Systems, Digital Image processing, Digital System Design, Computer Architecture.

Level in EQF EQF level 6

11 SEP 2017 – 10 JUN 2020 Wuhan, China

CONTROL SCIENCE & ENGINEERING Wuhan University of Science & Technology

Research: 1) Modelling of a Memristor by modified volterra series for parallel computing, 2) License plate recognition system (LPR) in machine vision.

Level in EQF EQF level 7

1 SEP 2022 – 22 SEP 2026 Jinan, China

CONTROL SCIENCE & ENGINEERING Shandong University

Experties in the field of signal processing, signal analysis, vibrational signals, rechargeable wireless sensor networks, battery swapping stations in renewable energy, sensors and instruments, deep learning algorithm optimization, digital twin, fault detection. PHM and fault generalization.

Level in EQF EQF level 8

PUBLICATIONS

2025

Multiscale Feature Fusion by Entropy-Augmented KCFP for Rolling Bearing Fault Diagnosis

Authors: Asim Shahzad, Fuzheng Liu, Mingshun Jiang, Faye Zhang, Zhaohui Gan | **Journal Name**: IEEE Sensors Journal | **Volume, Issue and Pages**: Volume: 25 Issue: 14, pages=26421-26431

2025

Physics-inspired deep learning network using dilated kronecker convolution for rotary machines under variable operating conditions

Authors: Asim Shahzad, Fuzheng Liu, Mingshun Jiang, Faye Zhang, Liying Ren | **Journal Name**: Journal of the Franklin Institute | **Volume, Issue and Pages**: volume = 362, number = 18, pages = 108219

2026

Frequency-aware transformer fusion for intelligent fault diagnosis of rolling bearings under variable operating conditions

Authors: Asim Shahzad; Anqi Li, Fuzheng Liu, Mingshun Jiang, Faye Zhang, Yingzhe Jia | **Journal Name**: Measurement Science and Technology | **Volume, Issue and Pages**: Volume 37, Number 1 | **Publisher**: IOP

2026

Thermal-aware multi-objective optimization of battery swapping station with operational cost and battery degradation considerations

Authors: Asif Ali, Zhizhen Liu, Asim Shahzad | **Journal Name**: Scientific Reports | **Publisher**: Nature

2026

Learning-based energy-aware mobile charging under stochastic energy dynamics in rechargeable sensor networks

Authors: Asim Shahzad, Hang Zhou, Fuzheng Liu, Asif Ali, Mingshun Jiang | **Journal Name**: Engineering Research Express | **Publisher**: IOP

2026

A MILP–GIS Siting-and-Sizing Framework for Renewable-Powered Battery Swapping Station Networks with Hourly Feasibility Screening

Authors: Asif Ali, Zhizhen Liu, and Asim Shahzad | **Journal Name**: Electronics | **Volume, Issue and Pages**: 15(19), 4480 | **Publisher**: MDPI

CONFERENCES AND SEMINARS

22 JUN 2026 – 24 JUN 2026 Jinan, PR China

International Conference on Precision Instruments and Optical Engineering

11 MAY 2014 Multan, Pakistan

Youth Leadership and Peace Conference

HONOURS AND AWARDS

1 SEP 2017

Hubei Provincial Fellowship for Masters Education, China – Wuhan University of Science and Technology

1 SEP 2022

Chinese Government Scholarship for PhD Education, China – Shandong University

15 NOV 2012

Shahbaz Sharif Youth Initiative – Government of Pakistan

NETWORKS AND MEMBERSHIPS

18 MAY 2017 – 31 DEC 2028 Pakistan

Pakistan Engineering Council

Registered Engineer from Pakistan Engineering Council.
Registration Number: COMP/14508

SOCIAL AND POLITICAL ACTIVITIES

24 JUN 2024 – 30 JUN 2024 Qingdao, China
Encounter SCODA, Beautiful Qingdao

Research experience and practices.

SKILLS

deep learning | Fault detection | Wireless & Sensor Networks | sensors | signal processing | vibrational analysis | levels of software testing

LANGUAGE SKILLS

Mother tongue(s): **URDU**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	C2	C2	C2	C2	C2
CHINESE	C1	C1	C1	C1	B1

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Badminton, Reading History Books

DRIVING LICENCE

Driving Licence: B | 14 Feb 2023 – 9 Dec 2028